Nicolas Flammarion
Optimization for Machine Learning — CS-439 - MA
20.06.2025 from 15h15 to 18h15
Duration : 180 minutes

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Student 1

SCIPER : **999999**

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- This is a closed book exam. No electronic devices of any kind.
- Place on your desk: your student ID, writing utensils, one double-sided A4 page cheat sheet if you have one; place all other personal items below your desk or on the side.
- You each have a different exam.
- This exam has many questions. We do *not* expect you to solve all of them even for the best grade.
- Only answers in this booklet count. No extra loose answer sheets. You can use the last two pages as scrap paper.
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choisir une réponse | select an answer Antwort auswählen	ne PAS choisir une réponse | NOT select an answer NICHT Antwort auswählen	Corriger une réponse | Correct an answer Antwort korrigieren
☒ ☑ ■	☐	☐ ☐

ce qu'il ne faut **PAS** faire | what should **NOT** be done | was man **NICHT** tun sollte

First part: multiple choice questions

For each question, mark the box corresponding to the correct answer. Each question has **exactly one correct answer**.

Question 1 (Gradient Descent) Let $f(x) = \frac{1}{2}\mathbf{x}^\top Q\mathbf{x}$, where $Q \in \mathbb{R}^{d\times d}$ is a positive-definite matrix. Let $\lambda_1 > \lambda_2 > \cdots > \lambda_d > 0$ denote the eigenvalues of Q. Consider applying gradient descent with a fixed step size γ. There exists a threshold $\tilde{\gamma} > 0$ such that for any $\gamma > \tilde{\gamma}$, the algorithm diverges for certain initialization points. Determine the smallest such $\tilde{\gamma}$.

- ☐ $\frac{1}{\sum_{i=1}^d \lambda_i}$
- ☐ $\frac{1}{\lambda_1}$
- ☐ $\frac{2d}{\sum_{i=1}^d \lambda_i}$
- ☐ $\frac{2}{\sum_{i=1}^d \lambda_i}$
- ☐ $\frac{2}{\lambda_d}$
- ☐ $\frac{2}{\lambda_1}$
- ☐ $\frac{1}{\lambda_d}$
- ☐ $\frac{d}{\sum_{i=1}^d \lambda_i}$

Question 2 (Gradient Descent) Consider the problem of estimating a fixed but unknown vector $\mathbf{x} \in \mathbb{R}^d$. We are given a dataset of T observations, $\mathcal{D} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T\}$. Each observation $\mathbf{y}_t$ is generated independently by the following process:

$$\mathbf{y}_t = \alpha_t\mathbf{x} + \beta_t\boldsymbol{\varepsilon}_t$$

where α_t, β_t are known non-zero scalar coefficients and the noise is drawn independently from each other and $\mathbf{x}$ as $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, I_d)$ where I_d is the d by d identity matrix.
For a dataset $\mathcal{D}$ and model parameters $\boldsymbol{\theta}$, the likelihood function $q(\mathcal{D}|\boldsymbol{\theta})$ represents the probability of observing the data given the parameters. The maximum likelihood estimator (MLE) is the value of $\boldsymbol{\theta}$ that maximizes this function:

$$\hat{\boldsymbol{\theta}}_{\mathrm{MLE}} = \arg\max_{\boldsymbol{\theta}} q(\mathcal{D}|\boldsymbol{\theta})$$

Recall that for $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I_d)$, the probability density function is given by $f(\mathbf{z}) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|\mathbf{z}-\boldsymbol{\mu}\|_2^2}{2\sigma^2}\right)$. Which of the following optimization problems correctly formulates the MLE for the unknown parameter $\boldsymbol{\theta} = \mathbf{x}$ given the entire dataset $\mathcal{D}$ and parameters $\{(\alpha_t, \beta_t)|t \in \{1, \ldots, T\}\}$?

- ☐ $\max_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \frac{1}{\beta_t^2}\|\mathbf{y}_t - \alpha_t\mathbf{x}\|_2^2$.
- ☐ $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \|\mathbf{y}_t - \alpha_t\mathbf{x}\|_2^2$.
- ☐ $\max_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \exp\left(-\frac{\|\mathbf{y}_t-\alpha_t\mathbf{x}\|_2^2}{2\beta_t^2}\right)$.
- ☐ $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \frac{1}{\beta_t^2}\|\mathbf{y}_t - \alpha_t\mathbf{x}\|_2^2$.
- ☐ $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \beta_t^2\|\mathbf{y}_t - \alpha_t\mathbf{x}\|_2^2$.
- ☐ $\min_{\mathbf{x}\in\mathbb{R}^d} \sum_{t=1}^T \frac{1}{\beta_t}\|\mathbf{y}_t - \alpha_t\mathbf{x}\|_1$.
- ☐ $\max_{\mathbf{x}\in\mathbb{R}^d} \log\left(\sum_{t=1}^T \frac{1}{\beta_t}\exp(-\|\mathbf{y}_t - \alpha_t\mathbf{x}\|_2^2)\right)$.

Question 3 (Convexity) Let $A \in \mathbb{R}^{m \times d}$ and $\mathbf{w} \in \mathbb{R}^m$. Define the mapping

$$f : \mathbb{R}^d \longrightarrow \mathbb{R}, \qquad f(\mathbf{x}) = \mathbf{w}^\top \sigma(A\mathbf{x}),$$

where the ReLU activation $\sigma : \mathbb{R}^m \to \mathbb{R}^m$ is applied coordinate-wise; that is, for $\mathbf{z} \in \mathbb{R}^m$,

$$\big[\sigma(\mathbf{z})\big]_i = \max\{0, z_i\}, \qquad i = 1, \ldots, m.$$

Thus the network consists of an input linear layer $\mathbf{x} \mapsto A\mathbf{x}$, the ReLU activation, and an output linear layer $\sigma \mapsto \mathbf{w}^\top \sigma$. Under which of the following conditions is the mapping $\mathbf{x} \mapsto f(\mathbf{x})$ **convex**?

- ☐ All entries of the output weights must be positive.
- ☐ The output weight matrix must be positive semi-definite.
- ☐ The input weight matrix must be positive definite.
- ☐ All entries of the output weights must be non-negative.
- ☐ All entries of the input weights must be positive.
- ☐ The input weight matrix must be positive semi-definite.
- ☐ The output weight matrix must be positive definite.
- ☐ All entries of the input weights must be non-negative.

Question 4 (Convexity) Let $f : \mathbb{R}^d \to \mathbb{R}$ be a convex function. Consider the hypercube C in $\mathrm{dom}(f)$:

$$C = [l_1, u_1] \times [l_2, u_2] \times \ldots \times [l_d, u_d], \quad \text{with } l_i < u_i, \forall i$$

Define the set of the 2^d vertices of C by

$$V = \big\{\mathbf{v} \in \mathbb{R}^d \;:\; v_i \in \{l_i, u_i\} \text{ for every } i\big\}.$$

Pick an arbitrary point $\mathbf{x} \in C$. Which of the following statements is **necessarily true**?

- ☐ The minimum of f over C is attained only at a single point in V.
- ☐ The maximum of f over C can be attained at one or more points in V.
- ☐ The maximum of f over C is attained only at a single point in V.
- ☐ The minimum of f over C can occur on the boundary of C without ever occurring at a point in V.
- ☐ The minimum of f over C can be attained at one or more points in V.
- ☐ The maximum of f over C can occur on the boundary of C without ever occurring at a point in V.

Question 5 (Projected Gradient Descent) Consider the following function $f : [1, 3] \to \mathbb{R}$ defined as $f(x) = x^2 - 4x + 3$. Note that the function is defined on the closed interval $\mathcal{X} = [1, 3]$. We minimize f with the projected gradient descent algorithm, which is defined as follows:

$$x_{k+1} = P_{\mathcal{X}}(x_k - \gamma \nabla f(x_k)),$$

where $P_{\mathcal{X}}$ is the projection operator onto the set $\mathcal{X}$, and $\gamma > 0$ is a fixed step size. Over the choice of initialization x_0 and step size γ, which of the following scenarios are **not possible**?

- ☐ $x_7 = 3$ and $x_{14} = 1$
- ☐ $x_7 = 1.2$ and $x_{14} = 3$ and $x_{21} = 1$
- ☐ $x_7 = 1.2$ and $x_{14} = 1.6$ and $x_{21} = 1.8$
- ☐ $x_7 = 1.5$ and $x_{14} = 2.5$
- ☐ $x_7 = 1.2$ and $x_{14} = 1.8$ and $x_{21} = 2.4$

Question 6 (Proximal Gradient Descent) Consider the functions $g : \mathbb{R}^d \to \mathbb{R}$ and $h : \mathbb{R}^d \to \mathbb{R}$ defined as follows:

$$g(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top A\mathbf{x}, \quad h(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top B\mathbf{x}$$

where A, B are positive-definite diagonal matrices. Let $f = g + h$. We are interested in minimizing f using the following algorithms:

PGD-1 : With a fixed step size $\gamma > 0$, the proximal gradient descent algorithm is defined as

$$\mathbf{x}_{t+1}^{\text{PGD-1}} = \text{Prox}_{h,\gamma}\left(\mathbf{x}_t^{\text{PGD-1}} - \gamma \nabla g(\mathbf{x}_t^{\text{PGD-1}})\right)$$

PGD-2 : With a fixed step size $\gamma > 0$, the second proximal gradient descent algorithm with the proximal orcale defined with g is defined as

$$\mathbf{x}_{t+1}^{\text{PGD-2}} = \text{Prox}_{g,\gamma}\left(\mathbf{x}_t^{\text{PGD-2}} - \gamma \nabla h(\mathbf{x}_t^{\text{PGD-2}})\right)$$

GD : With a fixed step size $\gamma > 0$, the gradient descent algorithm is defined as

$$\mathbf{x}_{t+1}^{\text{GD}} = \mathbf{x}_t^{\text{GD}} - \gamma \nabla f(\mathbf{x}_t^{\text{GD}}).$$

Recall the proximal operator $\text{Prox}_{g,\gamma}(\mathbf{z})$ is defined as

$$\text{Prox}_{g,\gamma}(\mathbf{z}) = \arg\min_{\mathbf{y}} \left\{ \frac{1}{2\gamma} ||\mathbf{y} - \mathbf{z}||^2 + g(\mathbf{y}) \right\}.$$

where $||\cdot||$ is the Euclidean norm. When the three above algorithms are initialized at $\mathbf{x}_0$ and are run with a same step size $0 < \gamma < \frac{1}{L}$, where L is the smoothness constant of f. For $t > 0$, which of the following statements is **always true** regarding the rate of convergence of function f?

- ☐ $f(\mathbf{x}_t^{\text{PGD-1}}) = f(\mathbf{x}_t^{\text{PGD-2}}) \leqslant f(\mathbf{x}_t^{\text{GD}})$
- ☐ $f(\mathbf{x}_t^{\text{GD}}) \leqslant f(\mathbf{x}_t^{\text{PGD-1}})$
- ☐ $f(\mathbf{x}_t^{\text{PGD-1}}) \leqslant f(\mathbf{x}_t^{\text{GD}})$ and $f(\mathbf{x}_t^{\text{PGD-2}}) \leqslant f(\mathbf{x}_t^{\text{GD}})$
- ☐ None of the remaining statements are always true.
- ☐ $f(\mathbf{x}_t^{\text{PGD-1}}) = f(\mathbf{x}_t^{\text{PGD-2}}) \geqslant f(\mathbf{x}_t^{\text{GD}})$

Question 7 (Stochastic Gradient Descent) Let $f : \mathbb{R} \to \mathbb{R}$ be *μ-strongly convex* and *L-smooth*, with $\mu > 0$ and $L > 0$. Let $\mathbf{x}^\star$ denote the unique minimizer of f, and define the minimum value as $f^\star = f(\mathbf{x}^\star)$. At each iteration $t = 0, 1, 2, \ldots$, we observe a stochastic gradient $\mathbf{g}_t$ satisfying:

$$\mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] = f'(\mathbf{x}_t), \qquad \mathbb{E}\big[(\mathbf{g}_t - f'(\mathbf{x}_t))^2 \mid \mathbf{x}_t\big] = \sigma^2, \quad \text{with } \sigma^2 > 0.$$

For a fixed step size $0 < \gamma < \frac{1}{L}$, consider the following updates:

$$\text{GD: } \mathbf{x}_{t+1} = \mathbf{x}_t - \gamma f'(\mathbf{x}_t), \qquad \text{SGD: } \mathbf{x}_{t+1} = \mathbf{x}_t - \gamma \mathbf{g}_t.$$

We say that a method "oscillates around $f^\star$ within range $\mathcal{O}(\xi)$" if there exists a $T_0 \in \mathbb{N}$ and a constant $C > 0$ that is independent of ξ such that:

$$\mathbb{E}[f(\mathbf{x}_t)] - f^\star \leq C\xi \quad \text{for all} \quad t \geq T_0.$$

Choose the statement that is **always true** under the conditions above:

- ☐ GD oscillates around $f^\star$ within $\mathcal{O}(\gamma\sigma^2)$ while SGD converges to $f^\star$.
- ☐ SGD oscillates around $f^\star$ within range $\mathcal{O}(\gamma\sigma^2)$, while GD is guaranteed to converge to $f^\star$.
- ☐ Both methods do not converge and oscillate around $f^\star$ within range $\mathcal{O}(\gamma\sigma^2)$.
- ☐ Both methods are guaranteed to converge to $f^\star$.

Question 8 (Non-convex Optimization) Consider the non-convex function $f : \mathbb{R} \to \mathbb{R}$ defined via the mapping $f(x) = -e^{-x^2}$. In order to minimize f, we run gradient descent (GD) on f with step size $0 < \gamma < \infty$ starting from some $x_0 \neq 0$. Which of the following is true?

- ☐ We have no convergence guarantees for GD since f is non-smooth.
- ☐ Even for small step sizes, there exist some points of initialization such that GD diverges to either $-\infty$ or $+\infty$ since $\lim_{x\to\pm\infty} \nabla f(x) = 0$.
- ☐ GD converges to 0 for step size $1/2$, but not for step size $\gamma = 2$.
- ☐ Depending on initialization and step-size, GD can converge to either one of the three points of inflection $\{0, \pm\sqrt{1/2}\}$ of f.
- ☐ None of the other options is true.
- ☐ GD converges to 0 for step sizes $\gamma = 1/2$ and $\gamma = 2$.

Question 9 (Polyak-Lojasiewicz Inequality) Assume $f : \mathbb{R}^d \to \mathbb{R}$ fulfils the Polyak-Lojasiewicz (PL) inequality. Which of the following functions are PL (with some arbitrary parameter $0 < \mu < +\infty$)?

- $h_1(\mathbf{x}) := f(A\mathbf{x})$, where A is a square invertible matrix.
- $h_2(\mathbf{x}) := [f(\mathbf{x})]^2$.
- $h_3(\mathbf{x}) := f(\mathbf{x}) + g(\mathbf{x})$ where g also satisfies PL.

- ☐ Only h_2.
- ☐ Only h_1 and h_3.
- ☐ None of the functions has the PL property.
- ☐ Only h_1.
- ☐ Only h_1 and h_2.
- ☐ Only h_2 and h_3.
- ☐ Only h_3.

Question 10 (Linear Minimization Oracles) We consider the computational complexity of linear minimization oracles (LMOs) for the constraint set $\mathcal{X}$ being unit radius $\mathcal{L}_p$ balls in $\mathbb{R}^d$.
Recall that $\mathcal{B}_p = \left\{ \mathbf{x} \in \mathbb{R}^d \;\middle|\; \left(\sum_{i=1}^d |x_i|^p\right)^{1/p} \leq 1 \right\}$, for $p \in \{1,2\}$ and $\mathcal{B}_\infty = \{\mathbf{x} \in \mathbb{R}^d \mid \max_{1\leq i \leq d} |x_i| \leq 1\}$.
Which of the following statements are the only true ones?

A) There exist LMOs for $\mathcal{B}_1, \mathcal{B}_2$, and $\mathcal{B}_\infty$ that can all be evaluated in $\mathcal{O}(d)$ time.

B) Any LMO for $\mathcal{B}_\infty$ has time complexity $\Omega(2^d)$ because $\mathcal{B}_\infty$ has 2^d extremal points.

C) There exist an LMO for $\mathcal{B}_2$ that has time complexity $\mathcal{O}(1)$, since it suffices for the LMO to simply rescale the gradient.

D) In terms of per-iteration cost, Frank-Wolfe is preferable over projected gradient descent for $\mathcal{X} = \mathcal{B}_1$.

E) In terms of per-iteration cost, Frank-Wolfe is preferable over projected gradient descent for $\mathcal{X} = \mathcal{B}_2$.

F) In terms of per-iteration cost, Frank-Wolfe is preferable over projected gradient descent for $\mathcal{X} = \mathcal{B}_\infty$.

- ☐ B), C), D) and E)
- ☐ A) and D)
- ☐ B), D) and E)
- ☐ C) and E)
- ☐ C), D) and E)
- ☐ A), D) and F)
- ☐ B) and E)
- ☐ A), C), E) and F)

Question 11 (Subgradients) Given a convex function $f : \mathrm{dom}(f) \to \mathbb{R}$, a point $\mathbf{x}_0 \in \mathrm{dom}(f) \subseteq \mathbb{R}^d$ and a subgradient $\mathbf{g} \in \partial f(\mathbf{x}_0)$, which of the following statements are true?

A) There exists a point $\mathbf{x}_1 \neq \mathbf{x}_0$ in the domain of f such that $\mathbf{0} \in \partial f(\mathbf{x}_1)$.

B) There exists a point $\mathbf{x}_2 \neq \mathbf{x}_0$ in the domain of f such that $\mathbf{g} \in \partial f(\mathbf{x}_2)$.

C) Given a point $\mathbf{x}_3 \neq \mathbf{x}_0$ in the domain of f such that $\|\mathbf{x}_3 - \mathbf{x}_0\| \leq R$, then $f(\mathbf{x}_3) - f(\mathbf{x}_0) \leq R\|\mathbf{g}\|$.

D) Given a subgradient $\mathbf{g}' \in \partial f(\mathbf{x}_0)$, then $\lambda\mathbf{g} + (1-\lambda)\mathbf{g}'$ for any $\lambda \in [0,1]$ is also a subgradient of f at $\mathbf{x}_0$.

E) Assuming f is differentiable at $\mathbf{x}_0$, the set $\{\lambda\mathbf{g} + (1-\lambda)\nabla f(\mathbf{x}_0) \mid \lambda \in [0,1]\}$ consists of subgradients and is not a singleton.

- ☐ C), D) and E)
- ☐ B)
- ☐ A), B) and C)
- ☐ B), C) and D)
- ☐ D)
- ☐ A), B) and E)
- ☐ C) and E)
- ☐ A) and B)
- ☐ D) and E)

Question 12 (Newton's method) Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as $f(x) = x^3/3 - x$. We run gradient descent with fixed step size $\gamma > 0$ and Newton's method both from some initial point $x_0 \in \mathbb{R}$. Which of the following statements are true?

A) Assume $x_0 \gg 0$, the gradient descent algorithm converges to a local minimum of f.

B) Regardless of the initial point x_0, the iterates of gradient descent with large enough step size γ diverges.

C) For any initial point $x_0 \neq 0$, Newton's method converges to a critical point of f.

D) For any initial point $x_0 \neq 0$, Newton's method has at most single iterate that is inside the interval $(-1, 1)$.

- ☐ A), B) and D)
- ☐ B) and D)
- ☐ B), C) and D)
- ☐ A), B) and C)
- ☐ A) and B)
- ☐ A) and D)
- ☐ A), C) and D)
- ☐ C) and D)
- ☐ A) and C)
- ☐ B) and D)

Second part: true/false questions

For each question, mark the box (without erasing) TRUE if the statement is **always true** and the box FALSE if it is **not always true** (i.e., it is sometimes false).

Question 13 (Gradient Descent) The direction of the gradient points to the steepest descent direction, which is the direction pointing to a local or global minimum.

☐ TRUE ☐ FALSE

Question 14 (Subgradients) Consider a function $f : \mathbb{R}^d \to \mathbb{R}$ such that for any $\mathbf{x} \in \mathbb{R}^d$ and for any $\mathbf{g} \in \partial f(\mathbf{x})$, $\|\mathbf{g}\| \leq B$. Then, f is Lipschitz continuous with constant $L = B$.

☐ TRUE ☐ FALSE

Question 15 (Lower Bound) There always exists a B-Lipschitz convex function $f : \mathbb{R}^d \to \mathbb{R}$ with the following property: For any (sub)gradient-based method initialized at $\mathbf{x}_0$ and run for $T \geq 1$ iterations, the objective error satisfies

$$f(\mathbf{x}_T) - f(\mathbf{x}^\star) \geq c\left(\frac{B}{\sqrt{T}}\right),$$

for some constant $c > 0$, where $\mathbf{x}^\star$ is the global minimum of f.

☐ TRUE ☐ FALSE

Question 16 (Convexity) Any univariate differentiable function with a convex gradient is convex.

☐ TRUE ☐ FALSE

Question 17 (Smooth Functions) Let f be a L-smooth convex function. Consider the gradient descent algorithm with stepsize $\gamma = \frac{1}{L}$ from an initial point $\mathbf{x}_0$ such that $\|\mathbf{x}_0 - \mathbf{x}^\star\| \leq R$, where $\mathbf{x}^\star$ is the global minimum of f. Then, after T iterations,

$$f\left(\frac{1}{T}\sum_{i=1}^{T}\mathbf{x}_i\right) - f(\mathbf{x}^\star) = \mathcal{O}(1/T).$$

☐ TRUE ☐ FALSE

Question 18 (Convexity) A convex function is continuous.

☐ TRUE ☐ FALSE

Question 19 (Subgradients) Consider a function $f : \mathbb{R}^d \to \mathbb{R}$ defined by

$$f(\mathbf{x}) = \max_{1 \leq j \leq d} x_j + \frac{1}{2}\|\mathbf{x} - \mathbf{a}\|^2,$$

for some $\mathbf{a} \in \mathbb{R}^d$. There exist a unique point $\mathbf{x}^\star$ such that $\mathbf{0} \in \partial f(\mathbf{x}^\star)$.

☐ TRUE ☐ FALSE

Question 20 (Quasi-Newton Methods) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. Then, the Quasi-Newton methods differ in their choice of $H_t \in \mathbb{R}^{d \times d}$ that verifies the following d-dimensional secant condition:

$$\nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_{t-1}) = H_t(\mathbf{x}_t - \mathbf{x}_{t-1}).$$

☐ TRUE ☐ FALSE

Question 21 (SGD) Averaging the stochastic gradients over a mini-batch of size m can reduce the variance by a factor of $\frac{1}{m^2}$ compared to the single-sample variance.

☐ TRUE ☐ FALSE

Question 22 (Convexity) Suppose there is a two-layer neural network: $f : \mathbb{R}^d \to \mathbb{R}, f(\mathbf{x}) = W_2 W_1 \mathbf{x}$, with two linear mappings W_1, W_2 and no activation function in between. The neural network is trained using a mean-squared error loss $\mathcal{L}$. $\mathcal{L}$ is convex in (W_1, W_2).

☐ TRUE ☐ FALSE

Question 23 (Newton's Method) Newton's method is invariant under any invertible affine transformation.

☐ TRUE ☐ FALSE

Question 24 (ClippedSGD) A clipped stochastic gradient is an unbiased estimator of the true gradient.

☐ TRUE ☐ FALSE

Third part, open questions

Answer in the empty space below. Your answer should be carefully justified, and all the steps of your argument should be discussed in details. Leave the check-boxes empty, they are used for the grading.

1 Affine Invariance of Frank-Wolfe

Question 25: (*3 points*) Show that the Frank-Wolfe algorithm is *affine invariant*, which formally means the following:

Let $\{\mathbf{x}_k\}$, $\mathbf{x}_k \in \mathbb{R}^d$, be the sequence of iterates generated by applying the Frank-Wolfe algorithm to the problem $\min_{\mathbf{x}\in\mathcal{C}} f(\mathbf{x})$ starting from $\mathbf{x}_0$. Let $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with A invertible. If the algorithm is now applied to the transformed problem $\min_{\mathbf{y}\in T(\mathcal{C})} f(T^{-1}(\mathbf{y}))$ starting from the transformed point $\mathbf{y}_0 = T(\mathbf{x}_0)$, the resulting sequence of iterates $\{\mathbf{y}_k\}$ will satisfy $\mathbf{y}_k = T(\mathbf{x}_k)$ for all $k \geq 0$.

☐0 ☐1 ☐2 ☐3

2 Cocoercivity

The goal of the exercise is to establish the cocoercivity inequality

$$\frac{1}{2L}\|\nabla f(\mathbf{z}) - \nabla f(\mathbf{y})\|^2 \leq f(\mathbf{z}) - f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{y} - \mathbf{z}\rangle \tag{Coco}$$

characterizing smooth convex functions.

Question 26: (*3 points*) Show that f is convex and L-smooth if and only if, $\forall(\mathbf{x}, \mathbf{y}, \mathbf{z})$

$$f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \leq f(\mathbf{z}) + \langle \nabla f(\mathbf{z}), \mathbf{x} - \mathbf{z}\rangle + \frac{L}{2}\|\mathbf{x} - \mathbf{z}\|^2. \tag{1}$$

☐ 0 ☐ 1 ☐ 2 ☐ 3

Question 27: (*3 points*) Show that f is convex and L-smooth if and only if, $\forall(\mathbf{y},\mathbf{z})$

$$0 \leq f(\mathbf{z}) - f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{y} - \mathbf{z} \rangle - \frac{1}{2L} \|\nabla f(\mathbf{z}) - \nabla f(\mathbf{y})\|^2 \,, \tag{2}$$

i.e.,

$$\frac{1}{2L} \|\nabla f(\mathbf{z}) - \nabla f(\mathbf{y})\|^2 \leq f(\mathbf{z}) - f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{y} - \mathbf{z} \rangle \,. \tag{3}$$

☐ 0 ☐ 1 ☐ 2 ☐ 3

Question 28: (*3 points*) Show that f is convex and L-smooth if and only if, $\forall(\mathbf{y},\mathbf{z})$

$$\frac{1}{L}\|\nabla f(\mathbf{z})-\nabla f(\mathbf{y})\|^2 \leq \langle \nabla f(\mathbf{y})-\nabla f(\mathbf{z}), \mathbf{y}-\mathbf{z}\rangle \,. \tag{4}$$

Hint: for the reverse direction, show that f satisfies for any $\boldsymbol{\eta},\boldsymbol{\theta}\in\mathbb{R}^d$, $\langle\boldsymbol{\eta}-\boldsymbol{\theta},\nabla f(\boldsymbol{\eta})-\nabla f(\boldsymbol{\theta})\rangle \geq 0$, iif f is convex. Consider $g(t)=f(\boldsymbol{\theta}+t(\boldsymbol{\eta}-\boldsymbol{\theta}))$. Show that $t\geq 0$, $g'(t)\geq g'(0)$ and prove $f(\boldsymbol{\eta})\geq f(\boldsymbol{\theta})+\langle\nabla f(\boldsymbol{\theta}),\boldsymbol{\eta}-\boldsymbol{\theta}\rangle$.

☐0 ☐1 ☐2 ☐3

Question 29: (*2 points*) Find a inequality similar to (Coco) characterizing smooth strongly convex functions.

☐0 ☐1 ☐2

Simplest proof of Nesterov Accelerated Gradient

Consider the problem of minimizing a L-smooth convex function $f : \mathbb{R}^d \to \mathbb{R}$. For this, we consider the Nesterov accelerated gradient method which is defined by the update rule

$$\begin{cases} \mathbf{y}_t &= \mathbf{x}_t + \beta_t(\mathbf{x}_t - \mathbf{x}_{t-1}) \\ \mathbf{x}_{t+1} &= \mathbf{y}_t - \frac{1}{L}\nabla f(\mathbf{y}_t) \end{cases} \tag{NAG}$$

where $(\beta_t)_{t\in\mathbb{N}}$ is called momentum parameter. For the sake of convenience, let $\mathbf{x}_{-1} = \mathbf{x}_0$ and for $t \geq 0$, define

$$V_t \triangleq \lambda_t^2(f(\mathbf{x}_t) - f^\star) + \frac{L}{2}\|\lambda_t(\mathbf{x}_t - \mathbf{x}^\star) + (1-\lambda_t)(\mathbf{x}_{t-1} - \mathbf{x}^\star)\|^2. \tag{5}$$

In the following questions, we will establish a rate of convergence for the NAG method using V_t.

Question 30: (*4 points*) Using (Coco), show that we have $V_{t+1} \leq V_t$ for any $t \geq 0$, with $\lambda_{t+1}^2 - \lambda_{t+1} = \lambda_t^2$ ($\lambda_0 = 0$) and $\beta_t = \frac{\lambda_t - 1}{\lambda_{t+1}}$.

☐0 ☐1 ☐2 ☐3 ☐4

Question 31: (*2 points*) Conclude by providing the convergence rate of the function value.

☐0 ☐1 ☐2

Question 32: (*2 points*) You have obtained from Question 30 that $V_{t+1} - V_t \leq -\Delta_t$ for a specific non-negative Δ_t to specify. Prove that the sequence $V_t + \sum_{s=0}^{t-1} \Delta_s$ is non-increasing.

☐0 ☐1 ☐2

Question 33: (*3 points*) Conclude on a convergence guarantee of the smallest observed gradient norm.

☐0 ☐1 ☐2 ☐3